Practice Questions

Direct Proofs:

Give a direct proof of the theorem "If n is an odd integer, then n^2 is odd."

Give a direct proof that if m and n are both perfect squares, then nm is also a perfect square. (An integer a is a perfect square if there is an integer b such that $a = b^2$.)

Use a direct proof to show that the sum of two odd integers is even.

Use a direct proof to show that the sum of two even integers is even.

Show that the square of an even number is an even number using a direct proof.

Show that the additive inverse, or negative, of an even number is an even number using a direct proof.

Prove that if $m + n$ and $n + p$ are even integers, where $m, n,$ and p are integers, then $m + p$ is even. What kind of proof did you use?

Use a direct proof to show that the product of two odd numbers is odd.

Use a direct proof to show that every odd integer is the difference of two squares.

Prove that the sum of two rational numbers is rational.

Proof by Contraposition:

Prove that if n is an integer and $3n + 2$ is odd, then n is odd.

Prove that if $n = ab$, where a and b are positive integers, then $a \leq \sqrt{n}$ or $b \leq \sqrt{n}$.

Prove that if n is an integer and n^2 is odd, then n is odd.

Proof by Contradiction:

Show that at least four of any 22 days must fall on the same day of the week.

Prove that $\sqrt{2}$ is irrational by giving a proof by contradiction.

Give a proof by contradiction of the theorem "If $3n + 2$ is odd, then n is odd."

Proofs of Equivalence:

Prove the theorem "If n is an integer, then n is odd if and only if n^2 is odd."

Show that these statements about the integer n are equivalent:

p_1: n is even.

p_2: $n - 1$ is odd.

p_3: n^2 is even.

Counterexamples:

Show that the statement “Every positive integer is the sum of the squares of two integers” is false.

Proof by Mathematical Induction:

Show that if n is a positive integer, then

$$1 + 2 + \cdots + n = \frac{n(n\ +\ 1)}{2}.$$

Conjecture a formula for the sum of the first n positive odd integers. Then prove your conjecture using mathematical induction.

Use mathematical induction to show that

$$1 + 2 + 2^2 + \cdots + 2^n = 2^{(n+1)} - 1$$

for all nonnegative integers n.

Use mathematical induction to prove this formula for the sum of a finite number of terms of a geometric progression with initial term a and common ratio r:

$$\sum_{j=0}^{n} ar^j = a + ar + ar^2 + \cdots + ar^n = \frac{ar^{n+1} - a}{r - 1} \qquad \text{when} \quad r \neq 1,$$

where n is a nonnegative integer.

Use mathematical induction to prove the inequality

$$n < 2^n$$

for all positive integers n.

Use mathematical induction to prove that $2^n < n!$ for every integer n with $n \geq 4$. (Note that this inequality is false for $n = 1,2, \text{and } 3$.)

Use mathematical induction to prove that $n^3 - n$ is divisible by 3 whenever n is a positive integer.

Use mathematical induction to prove that $7^{(n+2)} + 8^{(2n+1)}$ is divisible by 57 for every nonnegative integer n.

Use mathematical induction to show that if S is a finite set with n elements, where n is a nonnegative integer, then S has 2^n subsets.

An odd number of people stand in a yard at mutually distinct distances. At the same time each person throws a pie at their nearest neighbor, hitting this person. Use mathematical induction to show that there is at least one survivor, that is, at least one person who is not hit by a pie. (Note that this result is false when there are an even number of people.)